

Restaurant POS & Billing System

Project Summary

Project Title

Restaurant POS & Billing System

Objective

To modernize restaurant billing by automating tax and service charge calculations while ensuring real-time inventory synchronization to prevent stock-outs during peak hours.

Core Accomplishment

- Developed a high-speed, full-stack POS application featuring a dynamic "Digital Menu" and an automated checkout workflow.
- The system's standout feature is the **Manual Override Discount Engine**, which utilizes JavaScript prompt-sequences to allow cashiers to apply flexible discounts during the final payment stage without cluttering the primary user interface.

Technical Complexity

- **Automated Inventory Replenishment:** Implemented a "Smart Refill" trigger that monitors stock levels during every transaction and auto-replenishes items to a baseline of 50 units upon depletion.
- **Atomic Transaction Processing:** Utilized Django's transaction.atomic to ensure data integrity, preventing stock deductions if a payment or order creation fails.
- **Dynamic PDF & Excel Reporting:** Engineered a reporting engine using ReportLab and OpenPyxl to generate professional tax invoices and sales summary reports with auto-scaling column logic and visual styling.
- **Optimized UI/UX Workflow:** Reduced checkout friction by migrating coupon validation from a static input field to a conditional JavaScript workflow, improving operational speed for staff.

Tech Stack

- **Backend:** Python, Django (MTV Architecture)
- **Frontend:** HTML5, CSS3, JavaScript (Bootstrap 5, BI Icons)
- **Database:** MySQL / PostgreSQL / SQLite
- **Reporting:** ReportLab (PDF Engine), OpenPyxl (Excel Generation)
- **Environment Management:** Virtualenv (venv)